MIL-T-36820
12 April 1972

MILITARY SPECIFICATION

TOOTH DIE PIN PLACEMENT KIT

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a tooth die pin placement kit suitable for use in the placement and positioning of dowel pins in dental elastic impression material during pouring of the stone cast.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

FSC 6520

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STANDARDS

Military

MIL-STD-129 Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.

3.1.1 Harness. The harness (multi-colored plastic strip) shall be made of a vinyl compound.

3.1.2 Cap. Dowel cap shall be made of a vinyl plastisol compound.

3.1.3 T-pin. T-pin shall be fabricated of nickel-plated steel.

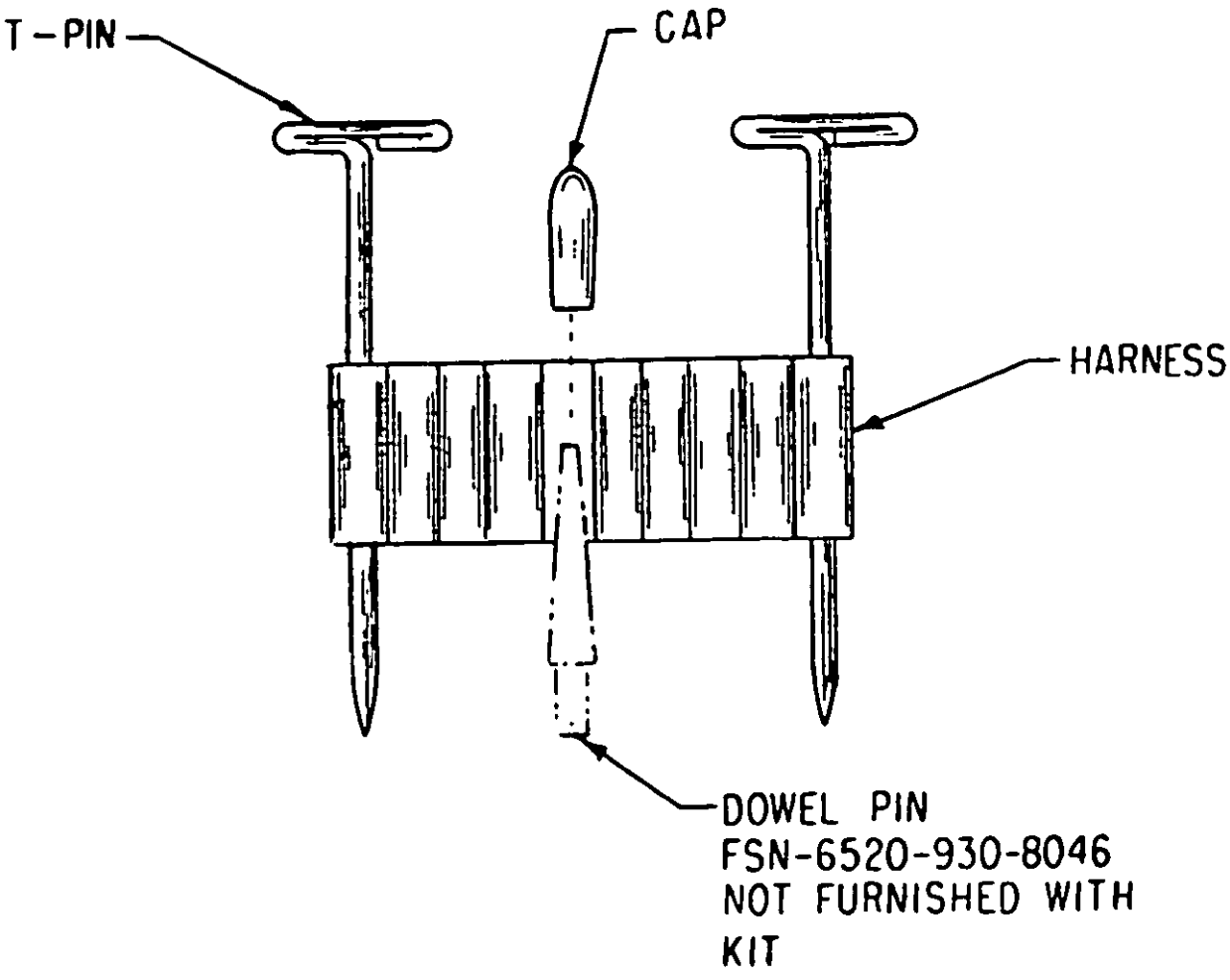
3.2 Figures. The figures forming a part of this specification are specification control figures. The supplier is responsible for preparing his own shop drawings. Where tolerances prescribed could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the tooth die pin placement kit. No deviation from prescribed dimensions or tolerances is permissible without prior approval of the contracting officer.

3.3 Style, design, and dimensions. Tooth die pin placement kit shall consist of the following items:

(a) Fifty (50) T-pins (min.)
(b) Fifty (50) harnesses (min.)
(c) Fifty (50) dowel caps (min.)
(d) Plastic immediate container

The kit shall be suitable for use in the placement and positioning of the dowel pin in various types of dental elastic impression material during pouring of the stone cast. All items in the kit shall conform to Figures 1 and 2 for style, design, and dimensions.

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ASSEMBLY

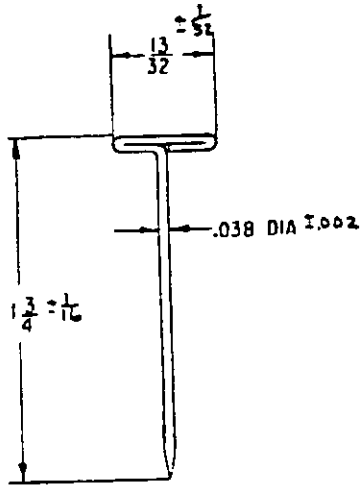
FIGURE 1

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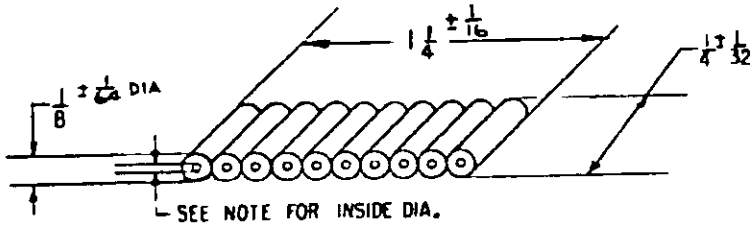


CAP B DOWEL PIN ASSEMBLY

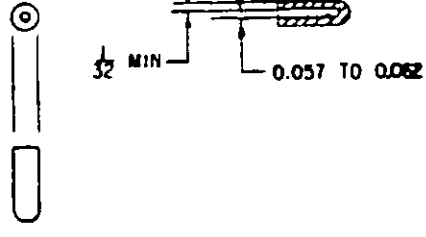
DOWEL PIN: FSN 6520-930-8046
NOT FURNISHED WITH KIT



T-PIN



HARNESS



CAP

NOTE: INSIDE DIAMETER OF HARNESS TUBES SHOULD BE LESS THAN 036 INCH TO INSURE THAT THE T-PIN WILL FIT SNUGLY IN TO IT, EXCEPT FOR END HARNESS TUBES IN WHICH DIAMETER MAY BE LARGER.

FIGURE 2

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3.3.1 T-pin. The T-pin shall be a straight steel pin with a cross bar used to position the harness over the proper area of impression material.

3.3.2 Harness. The harness (multi-colored plastic strip) shall consist of 10 hollow tubes of various translucent colors (including colorless) attached together to form a strip. Two colorless tubes or two tubes of the same color shall not appear in succession in the harness. The diameter of the tube shall be such that each tube shall accommodate the tapered end of the dowel pin, except for the end tubes in which the diameter may be slightly larger (see Fig. 2).

3.3.3 Cap. The cap shall be designed to fit onto the non-serrated end of the pin. The cap shall function as a marker for locating the end of the dowel pin in dental stone. The cap shall be of a bright opaque color.

3.4 Bonding strength of harness tubes. Harness tubes shall have sufficient bonding strength to withstand the test specified in 4.4.1.

3.5 Finish.

3.5.1 Pins. Nickel-plated steel for pins shall have a bright smooth finish, free of cracks, breaks, or burrs.

3.5.2 Harness and cap. Plastic for harness and cap shall have a smooth finish and be free of tears, breaks, or cracks.

3.6 Instructions for use. Instructions for use shall accompany each unit package.

3.7 Immediate container The kit, consisting of minimums of 50 T-pins, 50 harnesses (multi-colored plastic strips), and 50 dowel caps shall be furnished in a suitable, sturdy, plastic compartmented box, having 5 compartments; 4 compartments for the above-mentioned items in the kit, and the fifth compartment as a convenience to hold the dowel pins to be used in this technique.

3.8 Workmanship. Workmanship shall be first class throughout. Tooth die pin placement kits shall be free from defects which may detract from their appearance or impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables I and II (see 4.2.1 and 4.3) and 4.2.2.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table I. Unit of product for examination purposes shall be one kit (unit of issue).

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Table I. Inspection and Acceptable Quality Levels

Type defect	Inspection level	AQL (Percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests. Sampling for bond strength of harness tubes tests shall be conducted in accordance with MIL-STD-105 inspection level S-2, with an AQL of 2.5, except that the acceptance number shall be zero for all sample sizes. Unit of product for test purposes shall be one kit (unit of issue).

4.3 Examination. Placement kits shall be examined for defects, including, but not limited to, those listed in table II.

Table II. Classification of Defects

Categories	Defects
Major A	
101	Cracks, breaks, or burrs present in T-pins.
102	Cracks, breaks, or tears present in dowel cap or harness.
103	T-pin does not have a bright smooth surface.
104	Cap and harness do not have a smooth surface.
Major B	
151	Inside diameter of hole in harness not as specified on figure for all but end harness tubes in which diameter may be larger. 1/
152	Diameter of hole in cap not as specified. 2/
153	Wall thickness of cap not as specified.
154	Number of tubular elements in harness not as specified.
Minor A	
201	Harness not translucent and multi-colored.
202	Cap not bright color.
203	Immediate container not compartmentalized as specified in 3.7.

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Table II. Classification of Defects (cont'd.)

Categories	Defects
Minor B	
251	Dimensions, except for 151, 152, 153, and 154 not as specified.
252	Dimensions of immediate container not as specified.
253	Quantity not as specified.

1/ A suitable gage may be used, such as an 0.036 inch T-pin.
2/ A suitable gage may be used, such as a 1.2 mm. dowel pin.

4.4 Tests.

4.4.1 Bond strength of harness tubes. A T-pin shall be placed in each of the extreme end cylinders (tubes) of the harness. An 8-oz. load shall be securely attached with a strong thread or string to each end of one pin. The assembly shall then be suspended by holding the free ends of the unloaded pin by hand or in a jig to permit the harness and loaded pin to swing down around the held secured T-pin as a hinge. After 3 minutes under load, the harness shall be examined for breaks or ruptures in the bond between the cylinders.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. Each kit consisting of fifty T-pins, fifty strips, and fifty dowel caps, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table III shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table III. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	48 units

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5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. The fifty pins, strips, and caps along with instructions for use shall be packaged in a compartmented, plastic box of appropriate size. The box design shall include a clear plastic hinged cover and friction closure. In addition to the friction closure, the box shall be further secured by tape or other means so as to prevent spilling of contents under normal handling. At the option of the contractor, each plastic box may be inserted into a snug fitting paperboard box.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements as specified in Table II of PPP-B-636 shall apply for fiberboard boxes.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with 30.4 of PPP-B-636.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard boxes shall be closed by method II of PPP-B-636. Special requirements of Table I in PPP-B-636 shall apply for fiberboard boxes.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Federal Stock Number (FSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog: 6520-153-8472, PLACEMENT KIT, TOOTH DIE PIN.

6.3 Limitations. This specification does not cover all types, classes, grades, styles or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Military requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1309

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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